

Spectral Analysis of U.S. Equity Markets: Factor Structure, Sectoral Geometry, and Market Efficiency

Damon H. T. Cheung

University of Michigan, Ann Arbor, MI 48109

Abstract. We apply spectral and data-driven methods to daily returns of 20 large-cap U.S. equities (2020–2026) to ask: how many independent directions of variation does the market possess, and can that structure be exploited for prediction? Singular Value Decomposition (SVD) reveals only four statistically genuine factors; Classical Multidimensional Scaling (MDS) and Spectral Clustering recover the four known sectors without any labels. Dynamic Mode Decomposition (DMD) shows that returns are temporally unpredictable—a trained linear model underperforms a zero-change forecast, and a portfolio acting on its predictions loses -45.4% while the passive benchmark gains $+92.0\%$. Generalized Autoregressive Conditional Heteroskedasticity (GARCH) modelling shows the complementary result: while return *direction* is unpredictable, return *magnitude* (volatility) is forecastable, and a volatility-targeting portfolio achieves $+96.4\%$ with a higher Sharpe ratio. The market is low-dimensional across stocks yet unpredictable in time—the mathematical signature of market efficiency.

1 Introduction

The U.S. equity market is one of the most intensively studied dynamical systems in the world—and one of the most difficult to predict [2]. Yet most stocks move together: when the broad market falls, individual stocks fall with it regardless of their fundamentals. This co-movement raises a precise mathematical question: *how many independent directions of variation does the equity return matrix truly possess, and does this structure enable prediction?*

We address this question by applying SVD, Classical MDS, Spectral Clustering, Dynamic Mode Decomposition (DMD), and GARCH to daily returns of 20 large-cap U.S. stocks over 2020–2026. Our central finding resolves an apparent paradox: *the market is low-dimensional across stocks but unpredictable over time*. SVD reveals four genuine factors that spectral methods recover exactly from raw returns; DMD shows this structure cannot be extrapolated forward; GARCH shows that *volatility* can be forecast even when return direction cannot. The implication: mathematical methods can characterise market structure and manage risk, but cannot predict direction.

2 Data and Methods

Data. We use daily adjusted closing prices from Yahoo Finance for 20 large-cap U.S. stocks, five from each of four sectors—Technology (AAPL, MSFT, GOOGL, AMZN, META), Finance (JPM, BAC, GS, MS, C), Healthcare (JNJ, UNH, PFE, MRK, ABBV), and Energy (XOM, CVX, COP, SLB, EOG)—plus the S&P 500 index as an external benchmark. The period is 2 January 2020 to 24 April 2026 ($T = 1,585$ trading days, $N = 20$ stocks). We compute daily log-returns $r_{i,t} = \ln(p_{i,t}/p_{i,t-1})$ and standardise each stock to zero mean and unit variance, yielding the return matrix $\mathbf{X} \in \mathbb{R}^{T \times N}$.

SVD and Random-Matrix Baseline. We compute the thin SVD

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top, \quad \mathbf{\Sigma} = \text{diag}(\sigma_1 \geq \dots \geq \sigma_N \geq 0), \quad (1)$$

whose right singular vectors (columns of $\mathbf{V}$) are the principal components; the k -th explains variance fraction $\sigma_k^2 / \sum_j \sigma_j^2$. To separate genuine factors from finite-sample noise we apply the Marčenko–Pastur law [5]: for an i.i.d. Gaussian matrix with aspect ratio $q = N/T$, bulk eigenvalues are bounded by

$$\lambda_{\pm} = (1 \pm \sqrt{q})^2; \quad (2)$$

eigenvalues above λ_+ signal statistically genuine structure.

Correlation, MDS, and Spectral Clustering. We form the 20×20 Pearson correlation matrix $\mathbf{C}$, where

$$\rho_{ij} = \frac{\sum_t (r_{i,t} - \bar{r}_i)(r_{j,t} - \bar{r}_j)}{\sqrt{\sum_t (r_{i,t} - \bar{r}_i)^2 \cdot \sum_t (r_{j,t} - \bar{r}_j)^2}} \quad (3)$$

is the pairwise return correlation (which simplifies to $\rho_{ij} = \frac{1}{T-1} \sum_t r_{i,t} r_{j,t}$ after standardisation). Classical MDS [4] embeds stocks in $\mathbb{R}^2$ via the correlation distance $d_{ij} = \sqrt{2(1 - \rho_{ij})}$. Spectral clustering [6] with $k = 4$ (motivated by the four clusters visible in the MDS embedding) uses $\frac{1}{2}(\mathbf{C} + \mathbf{I})$ as an affinity matrix; we evaluate it with the Adjusted Rand Index (ARI; 1 = perfect agreement, 0 = chance).

Dynamic Mode Decomposition and Portfolio Test. DMD finds the best rank- r linear map satisfying $\mathbf{x}_{t+1} \approx \mathbf{A}\mathbf{x}_t$ [4]. Given snapshot matrices $\mathbf{X}_1, \mathbf{X}_2 \in \mathbb{R}^{N \times (T-1)}$ and a rank- r SVD $\mathbf{X}_1 = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^\top$, the reduced operator and its full-space reconstruction are

$$\tilde{\mathbf{A}} = \mathbf{U}_r^\top \mathbf{X}_2 \mathbf{V}_r \mathbf{\Sigma}_r^{-1}, \quad \mathbf{A} = \mathbf{U}_r \tilde{\mathbf{A}} \mathbf{U}_r^\top. \quad (4)$$

We set $r = N = 20$ (full rank, giving DMD maximum information); restricting to $r = 4$ (the M-P genuine factors) yields similar conclusions but less severe overfitting. We train on 2020–2022 (755 days) and test on 2023–2026 (830 days). Predictability is measured by the RMSE ratio $\text{RMSE}_{\hat{A}}/\text{RMSE}_0$ (ratio > 1 : the model is worse than predicting no change), and by a dollar-neutral long-short portfolio

$$r_t^{\text{port}} = \frac{1}{10} \sum_{i \in \mathcal{L}_t} r_{i,t} - \frac{1}{10} \sum_{i \in \mathcal{S}_t} r_{i,t}, \quad (5)$$

where $\mathcal{L}_t$ ($\mathcal{S}_t$) is the 10 highest (lowest) predicted-return stocks on day t , tracked against an equal-weight buy-and-hold benchmark.

GARCH Volatility Modelling and Volatility-Targeting Portfolio. Having shown that DMD cannot predict return *direction*, we ask whether return *magnitude* (volatility) is forecastable. We fit a GARCH(1,1) model [1] to S&P 500 returns:

$$r_t = \varepsilon_t \sqrt{h_t}, \quad h_t = \omega + \alpha r_{t-1}^2 + \beta h_{t-1}, \quad (6)$$

where h_t is the conditional variance, α captures the immediate shock impact, and β governs persistence. We evaluate the one-step-ahead volatility forecast on the 2023–2026 test set against the unconditional-mean baseline. We also construct a *volatility-targeting* portfolio whose daily position size $w_t = \min(\sigma^*/\hat{\sigma}_t, 2)$ scales the equal-weight portfolio to a target daily volatility $\sigma^* = 1\%$, capping leverage at $2\times$. Performance throughout is summarised by the annualised Sharpe ratio $\text{SR} = (\bar{r}_d/\hat{\sigma}_d)\sqrt{252}$. As a robustness check we also fit GARCH(1,1) to each of the 20 stocks individually, isolating idiosyncratic from systematic variance.

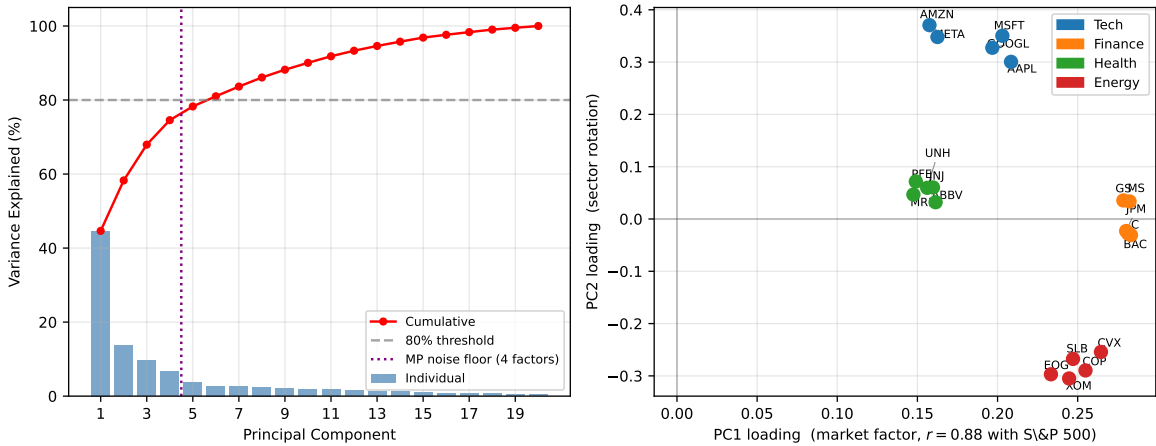


Fig. 1. SVD of the standardised return matrix $\mathbf{X}$. (a) Scree plot: the dotted vertical line marks the Marchenko–Pastur noise floor; only four eigenvalues exceed it. (b) Stock loadings on PC1 vs. PC2. Every stock has a positive PC1 loading (the market factor); PC2 separates Energy and Finance from Technology.

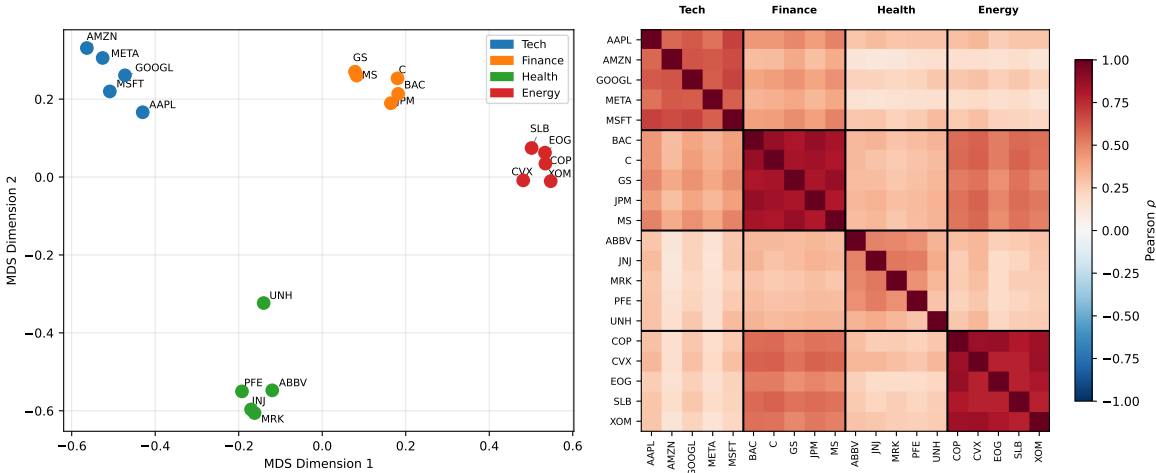


Fig. 2. (a) Classical MDS embedding: stocks cluster by sector with no label information supplied. (b) Pearson correlation matrix sorted by sector. Within-sector blocks are warm (high ρ_{ij}); cross-sector blocks are cool.

3 Results

3.1 The Market Is Low-Dimensional: Four Genuine Factors

PC1 captures 44.6% of total variance; the top three PCs together explain 67.9%. Crucially, the Marchenko–Pastur bound $\lambda_+ = 1.24$ is exceeded by exactly *four* eigenvalues (Fig. 1a)—all remaining 16 are statistically indistinguishable from random noise. The market is genuinely low-dimensional: a 20-stock universe reduces to four meaningful directions.

Figure 1b shows what those directions are. Every stock has a positive PC1 loading, confirming a single market-wide co-movement factor ($r = 0.883$ with the S&P 500 index)—the market factor of (author?) [3]. PC2 then separates Energy and Finance from Technology, hinting that the remaining factors capture sector-specific variation.

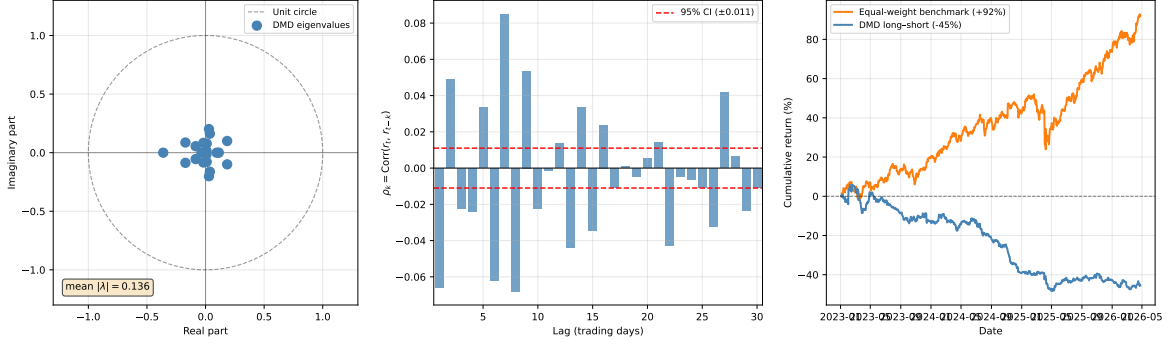


Fig. 3. (a) DMD eigenvalue spectrum (trained 2020–2022, tested 2023–2026). All eigenvalues lie well inside the unit circle (mean $|\lambda| = 0.136$), indicating modes that decay within one trading day. (b) Lag- k autocorrelation $\rho_k = \text{Corr}(r_t, r_{t-k})$: nearly every bar is within the 95% white-noise band (red dashes). (c) The DMD long-short portfolio loses -45.4% over the test period while the equal-weight benchmark gains $+92.0\%$.

3.2 Sector Structure Is Perfectly Recoverable Without Labels

The MDS embedding (Fig. 2a) translates correlation geometry into Euclidean space: Technology clusters tightly in the upper region, Energy to the lower left, with Finance and Healthcare in between. No sector labels were provided—the geometry emerges entirely from return correlations.

Spectral clustering with $k = 4$ reproduces the known sector partition exactly (ARI = 1.00), supported by the block-diagonal structure of the correlation heatmap (Fig. 2b). Within-sector correlations are consistently higher than cross-sector ones; Technology–Technology pairs are the most tightly linked, while Energy–Technology pairs are the most decorrelated.

What we learned: the four-factor structure found by SVD maps directly onto the four economic sectors. The market is not just low-dimensional—it is *interpretable* low-dimensional.

3.3 Returns Are Temporally Unpredictable: DMD Fails

Having established that the market’s cross-sectional structure is low-dimensional, we now ask whether that structure can be extrapolated forward in time. DMD finds the answer in its eigenvalues: all 20 lie well inside the unit circle (mean $|\lambda| = 0.136$, Fig. 3a). In systems with persistent dynamics—such as fluid oscillations—DMD eigenvalues sit *on* the unit circle and encode stable modes that can be tracked. Here every mode decays within a single trading day; there is no dynamical backbone to extrapolate.

This failure is confirmed statistically and economically. Statistically, the lag- k autocorrelation $\rho_k = \text{Corr}(r_t, r_{t-k})$ measures whether today’s return predicts tomorrow’s: values near zero for all $k \geq 1$ indicate white noise (Fig. 3b). The only exception is a small negative $\rho_1 = -0.066$, a well-known microstructure artefact (bid-ask bounce) that vanishes after transaction costs. Economically, a DMD model trained on 2020–2022 achieves an out-of-sample RMSE of 0.762 vs. the zero-change baseline of 0.735 (ratio 1.04)—the trained model is *worse* than predicting no change. A portfolio acting on these predictions loses -45.4% over the 2023–2026 test period, while a passive equal-weight portfolio gains $+92.0\%$ (Fig. 3c, annualised Sharpe = -1.23).

What we learned: the market’s stable spatial structure does not imply temporal predictability. Any linear pattern in returns is arbitrated away before it can be exploited—the mathematical signature of the Efficient Market Hypothesis [2].

3.4 Volatility Is Predictable: GARCH Succeeds Where DMD Fails

The DMD result establishes that the first moment of returns (expected value) is unpredictable. We now ask whether the second moment (variance) is equally featureless. Fitting GARCH(1,1) to S&P 500 returns yields $\hat{\alpha} = 0.158$ and $\hat{\beta} = 0.804$, giving persistence $\hat{\alpha} + \hat{\beta} = 0.962$. This near-unity persistence

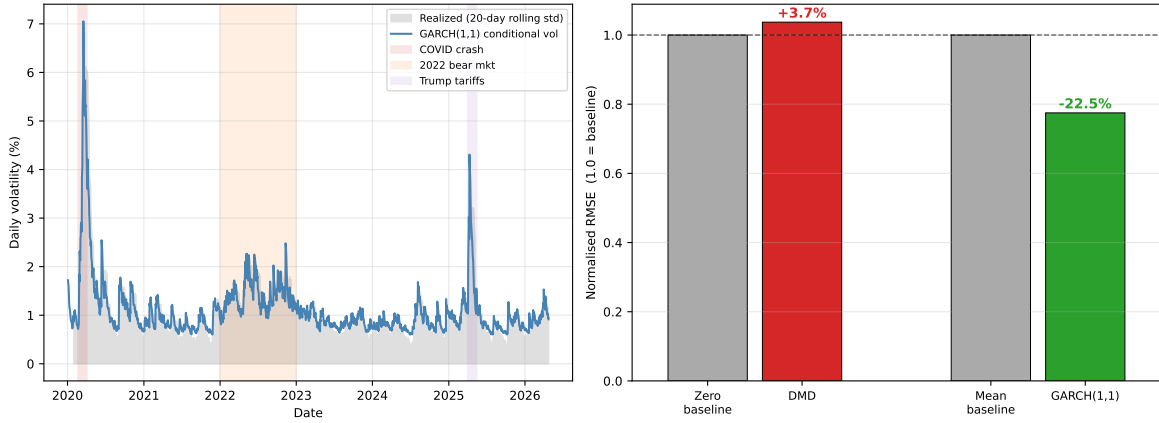


Fig. 4. (a) GARCH(1,1) conditional volatility vs. realised absolute returns (S&P 500). The model tracks the COVID-19 crash (Mar. 2020) and the 2022 rate-hike sell-off. **(b)** Normalised RMSE: DMD predicts *returns* worse than the zero baseline (ratio > 1); GARCH predicts *volatility* substantially better than the historical mean (ratio < 1).

means a large volatility shock today still echoes strongly tomorrow—volatility is long-lived and therefore forecastable. The model achieves a 22.5% reduction in RMSE on the held-out 2023–2026 period relative to predicting the historical mean (Fig. 4b). Applied stock-by-stock to the 20 individual equities, GARCH barely improves on the unconditional mean (8 of 20 stocks, mean RMSE ratio 1.005): the forecastable component is *systematic* market volatility (captured by the S&P 500 index), not idiosyncratic variance. In portfolio terms, a GARCH volatility-targeting strategy (+96.4%, Sharpe = 1.44) modestly outperforms the passive benchmark (+92.0%, Sharpe = 1.43) by scaling exposure up during calm periods and down during high-volatility episodes—contrasting sharply with the DMD long-short strategy (−45.4%, Sharpe = −1.23) that catastrophically underperforms the same benchmark.

What we learned: the EMH constrains the predictability of return *direction* (first moment), not return *magnitude* (second moment). Mathematical methods can manage risk but not manufacture alpha.

4 Conclusion

So what? We set out to ask whether the U.S. equity market is low-dimensional, and whether that structure enables prediction. The answer is: *yes to the first, no—and yes—to the second, depending on what you try to predict.*

Across stocks, the market is organised into exactly four statistically genuine factors that align perfectly with the four economic sectors. This structure is stable, interpretable, and recoverable without labels—a mathematical counterpart of the Fama–French factor model [3].

Over time, return *direction* is not predictable. DMD’s eigenvalues cluster at the origin rather than on the unit circle, and a portfolio trained on its predictions loses 45.4% while the market gains 92.0%—the mathematical signature of the Efficient Market Hypothesis [2]: any learnable pattern in prices is immediately arbitrated away. Return *magnitude*, however, is predictable: GARCH captures volatility persistence ($\hat{\alpha} + \hat{\beta} = 0.962$), reduces forecasting error by 22.5%, and enables a risk-adjusted strategy that outperforms the benchmark.

The take-away is precise: invest in risk management, not in return-prediction algorithms—spectral methods expose the boundary between what can and cannot be predicted with mathematical clarity.

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